

Wilcoxon test example

13 data points: 14; -4; -6; -2; -1; 18; 0; 6; 12; 8;
-4; 13; 7

Hypothesis

H_0 : median $m=0$
H_1 : $m > 0$

• delete observation 0 $\rightarrow n=12$

X_i	14	-4	-6	-2	-1	18	6	12	8	-4	13	7
$ X_i - m_0 $	14	4	6	2	1	18	6	12	8	4	13	7
rank	11	3 or 4 ↓	5 or 6 ↓	2	1	12	5 or 6 ↓	9	8	3 or 4 ↓	10	7
	11	3.5	5.5	2	1	12	5.5	9	8	3.5	10	7
+ signs	11	-3.5	-5.5	-2	-1	12	5.5	9	8	-3.5	10	7

$$W = 11 + (-3.5) + (-5.5) - 2 - 1 + 12 + 5.5 + 9 + 8 - 3.5 + 10 + 7 = \boxed{47}$$

When do we reject?

If $m > 0$, we have more positive values in the sum
("too many") $\rightarrow$ crit. region $C = \{W \geq c\}$

$$p\text{-value} = P(W \geq 47) =$$

"more extreme than observed"

$$P\text{-value} = P\left(Z \geq \frac{47-0}{\sqrt{\frac{n(n+1)(2n+1)}{6}}}\right) = P\left(Z \geq \frac{47}{25.5}\right) =$$

(n=12)

$$= P(Z \geq 1.84) \approx 0.033$$

So, we reject H_0 on ^{all} significance levels above 0.033,
for example, $\boxed{\alpha = 10\% \Rightarrow \text{reject } H_0}$

But $\boxed{\alpha = 3\% \Rightarrow \text{accept } H_0}$, since $p\text{-value} > \alpha$
 $0.033 > 0.03$

We can also find critical region for $\alpha = 0.1$

$$P(W \geq c) = 0.1 \rightarrow \text{find } c$$

$$P(W \geq c) = P\left(Z \geq \frac{c}{25.5}\right) \Rightarrow \frac{c}{25.5} = z_{0.1} = 1.282$$

$$\Rightarrow c = 32.69$$

$C = [32.69, \infty)$ critical region for $\alpha = 10\%$.

Since $47 \in C$, we reject H_0 on significance level 10%.